KCE PATHWAYS · GRADE 3 · UNIT 3: MARKET DAYS · WEEK 14 · MATH

Saturday at the Community Garden — Teacher Lesson Plan & Answer Key

Learning objectives: Students find the perimeter of polygons by adding side lengths, find an unknown side length when the perimeter is known, and decide whether a real-world job needs perimeter (feet) or area (square feet). They show that rectangles can have the same perimeter and different areas, or the same area and different perimeters.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: student types first and last name so results go to KingCretot Experience.
2. Warm-Up: Area Review (12 pts, Week 13 skill): counting unit squares, rows × columns, and area of a shape made of two rectangles. If a student misses two of three, review rows × columns before moving on.
3. Picture It: Saturday at the Community Garden. Introduces the Oak Street Community Garden in Atlanta, Ms. Washington, Jalen, Priya and Mateo, plus Atlanta garden facts. Ask: which garden jobs go around the edge, and which cover the inside?
4. Word Wrangler: 6 vocab cards (perimeter, area, side length, polygon, square unit, rectangle). Share the memory trick: perimeter has “rim” in it.
5. Pencil Break (Pre): student completes the Pre side of the printed sheet before any teaching. Do not help yet; it shows what they already know.
6. Perimeter: The Distance Around (12 pts). Tip: have the student trace each side with a finger and say its length while adding, so no side is skipped or counted twice.
7. Finding a Missing Side (12 pts). Tip: add the known sides, then subtract from the perimeter; for rectangles, take away both long sides first, then split what is left into two equal short sides. Includes two-step items.
8. Area or Perimeter? (12 pts). Tip: ask “Is the job around the edge or covering the inside?” Insist on units: feet for perimeter, square feet for area.
9. Same Perimeter, Different Area (12 pts). Tip: if you have grid paper or square tiles, let the student build 1 by 5, 2 by 4 and 3 by 3 with a 12-unit string around each.
10. Bonus Practice: The Path Border (XP only).
11. Match-Up Round (10 pts, 2 per first-try match).
12. Fix the Sentence (10 pts): typed answers are checked exactly (capital letters, end mark, list commas). Both serial-comma and no-serial-comma list forms are accepted.
13. Speed Arena and Keyboard Corral (XP only).
14. Boss Battle: The Big Garden Day (20 pts): two-step perimeter or area problem, which-explanation-is-correct, missing side, find-the-mistake.
15. Pencil Break (Post): student completes the Post side of the sheet and the star challenge if time allows.
16. Results: score of 85 or higher shows a custom Garden Hero congratulations with confetti; below 85 shows encouragement and a fresh question set on the next try (sets rotate 1, 2, 3).

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: AREA REVIEW (WEEK 13)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Each small square is 1 square unit. How many square units cover this rectangle?	**15 square units**	Count the squares, or multiply rows × columns: 3 × 5 = 15 square units.
w2_s0	Set 1: A rectangle has 4 rows of square tiles with 7 tiles in each row. Which shows its area?	**4 × 7 = 28 square units**	Area is rows × columns: 4 rows of 7 is 4 × 7 = 28 square units.
w3_s0	Set 1: This L-shaped bed is split into rectangle A and rectangle B. Each small square is 1 square unit. What is the total area?	**16 square units**	A: 3 rows × 4 columns = 12 square units. B: 2 rows × 2 columns = 4 square units. 12 + 4 = 16 square units.
w1_s1	Set 2: A garden mat has 5 rows of tiles with 5 tiles in each row. Each tile is 1 square foot. What is the area of the mat in square feet?	**25 square feet**	5 rows of 5 is 5 × 5 = 25 square feet.
w2_s1	Set 2: What does area tell you about a flat shape?	**How many unit squares cover it**	Area is the number of square units that cover a flat shape.
w3_s1	Set 2: This L-shaped bed is split into rectangle A and rectangle B. Each small square is 1 square unit. What is the total area?	**18 square units**	A: 2 rows × 6 columns = 12 square units. B: 3 rows × 2 columns = 6 square units. 12 + 6 = 18 square units.
w1_s2	Set 3: Each small square is 1 square unit. How many square units cover this rectangle?	**24 square units**	Multiply rows × columns: 4 × 6 = 24 square units.
w2_s2	Set 3: Find the mistake. Jalen says a rectangle with 3 rows of 8 squares has an area of 11 square units. What went wrong?	**He added 3 + 8. He should multiply: 3 × 8 = 24 square units.**	Area is rows × columns, so 3 × 8 = 24 square units, not 3 + 8.
w3_s2	Set 3: This L-shaped bed is split into rectangle A and rectangle B. Each small square is 1 square unit. What is the total area?	**17 square units**	A: 3 rows × 3 columns = 9 square units. B: 2 rows × 4 columns = 8 square units. 9 + 8 = 17 square units.

PERIMETER: THE DISTANCE AROUND

ID	Question	Answer	Why (explain to student)
p1a_s0	Set 1: The flower bed is 5 feet long and 2 feet wide. What is its perimeter in feet?	**14 feet**	Add all four sides: 5 + 2 + 5 + 2 = 14 feet.
p1b_s0	Set 1: The herb bed is a triangle. Its sides are 4 feet, 5 feet, and 6 feet long. What is its perimeter in feet?	**15 feet**	Add all three sides: 4 + 5 + 6 = 15 feet.
p1c_s0	Set 1: Find the mistake. Mateo walked around a triangle bed with sides of 3, 3, and 4 feet. He wrote 3 + 4 = 7 feet. What went wrong?	**He skipped a side. 3 + 3 + 4 = 10 feet.**	A triangle has three sides, so add all three: 3 + 3 + 4 = 10 feet.
p1a_s1	Set 2: A long bed is 8 feet long and 3 feet wide. What is its perimeter in feet?	**22 feet**	Add all four sides: 8 + 3 + 8 + 3 = 22 feet.
p1b_s1	Set 2: A five-sided bed has sides of 6, 2, 4, 4, and 2 feet. What is its perimeter in feet?	**18 feet**	Add all five sides: 6 + 2 + 4 + 4 + 2 = 18 feet.
p1c_s1	Set 2: Which explanation is correct? How do you find the perimeter of a shape?	**Add the lengths of all the sides.**	Perimeter is the distance around, so you add the lengths of all the sides.

ID	Question	Answer	Why (explain to student)
p1a_s2	Set 3: A square bed has 4 sides, and each side is 7 feet long. What is its perimeter in feet?	**28 feet**	7 + 7 + 7 + 7 = 28 feet. That is the same as 4 × 7 = 28.
p1b_s2	Set 3: Only two sides are labeled, but opposite sides of a rectangle match. The bed is 9 feet long and 4 feet wide. What is its perimeter in feet?	**26 feet**	The sides are 9, 4, 9, and 4 feet. 9 + 4 + 9 + 4 = 26 feet.
p1c_s2	Set 3: Find the mistake. Priya says the perimeter of a bed that is 7 feet long and 2 feet wide is 9 feet. What went wrong?	**She only added two sides. 7 + 2 + 7 + 2 = 18 feet.**	A rectangle has four sides. Add all four: 7 + 2 + 7 + 2 = 18 feet.

FINDING A MISSING SIDE

ID	Question	Answer	Why (explain to student)
u1a_s0	Set 1: This garden bed has a perimeter of 17 feet. Three sides are 5 feet, 3 feet, and 4 feet. How long is the missing side, in feet?	**5 feet**	Add the sides you know: 5 + 3 + 4 = 12. Then subtract: 17 − 12 = 5 feet.
u1b_s0	Set 1: A square bed has a perimeter of 12 feet. How long is each side?	**3 feet**	A square has 4 equal sides. 3 + 3 + 3 + 3 = 12, so each side is 3 feet.
u1c_s0	Set 1: Two steps! A rectangle bed has a perimeter of 22 feet. It is 7 feet long. How wide is it, in feet?	**4 feet**	Step 1: the two long sides are 7 + 7 = 14 feet. Step 2: 22 − 14 = 8 feet for the two short sides, and 4 + 4 = 8. The bed is 4 feet wide.
u1a_s1	Set 2: The triangle bed has a perimeter of 16 feet. Two sides are each 6 feet long. How long is the third side, in feet?	**4 feet**	Add the sides you know: 6 + 6 = 12. Then subtract: 16 − 12 = 4 feet.
u1b_s1	Set 2: Find the mistake. A bed has a perimeter of 20 feet. Three of its sides are 6, 4, and 6 feet. Jalen says the missing side is 16 feet. What went wrong?	**He forgot to subtract. 6 + 4 + 6 = 16, and 20 − 16 = 4 feet.**	16 feet is the total of the three sides he knows. Subtract it from the perimeter: 20 − 16 = 4 feet.
u1c_s1	Set 2: Two steps! A rectangle bed has a perimeter of 26 feet. It is 8 feet long. How wide is it, in feet?	**5 feet**	Step 1: the two long sides are 8 + 8 = 16 feet. Step 2: 26 − 16 = 10 feet for the two short sides, and 5 + 5 = 10. The bed is 5 feet wide.
u1a_s2	Set 3: A five-sided bed has a perimeter of 15 feet. The bottom side is missing. The other four sides are 2, 3, 3, and 2 feet. How long is the bottom side, in feet?	**5 feet**	Add the sides you know: 2 + 3 + 3 + 2 = 10. Then subtract: 15 − 10 = 5 feet.
u1b_s2	Set 3: Which explanation is correct? A rectangle has a perimeter of 20 feet and is 6 feet long. How can you find its width?	**Subtract 6 + 6 from 20 to get 8. Split 8 into two equal sides: 4 feet.**	The two long sides use 6 + 6 = 12 feet. 20 − 12 = 8 feet is left for two equal short sides, so each is 4 feet.
u1c_s2	Set 3: The fence around a square garden plot is 32 feet long. How long is each side, in feet?	**8 feet**	A square has 4 equal sides. 8 + 8 + 8 + 8 = 32, so each side is 8 feet. You can also think 32 ÷ 4 = 8.

AREA OR PERIMETER?

ID	Question	Answer	Why (explain to student)
c1a_s0	Set 1: Jalen wants to put a fence around the pepper bed. Should he find the area or the perimeter?	**The perimeter**	A fence goes around the edge, so Jalen needs the perimeter.
c1b_s0	Set 1: The pepper bed is 7 feet long and 2 feet wide. What is its area in square feet?	**14 square feet**	Area is rows × columns: 2 rows of 7 is 2 × 7 = 14 square feet.
c1c_s0	Set 1: What is the perimeter of the same pepper bed, in feet?	**18 feet**	Add all four sides: 7 + 2 + 7 + 2 = 18 feet.
c1a_s1	Set 2: Priya wants to cover the whole lettuce bed with straw. Should she find the area or the perimeter?	**The area**	Straw covers the inside of the bed, so Priya needs the area.
c1b_s1	Set 2: The lettuce bed is 6 feet long and 3 feet wide. Mateo puts a wooden border around it. How many feet of border does he need?	**18 feet**	A border goes around, so find the perimeter: 6 + 3 + 6 + 3 = 18 feet.
c1c_s1	Set 2: Find the mistake. Jalen says the area of a bed that is 4 feet long and 2 feet wide is 12 square feet. What went wrong?	**He found the perimeter. The area is 4 × 2 = 8 square feet.**	4 + 2 + 4 + 2 = 12 is the perimeter in feet. The area is 4 × 2 = 8 square feet.
c1a_s2	Set 3: Which job needs the area of a garden bed?	**Covering the bed with a layer of soil**	Soil covers the inside of the bed, so that job needs the area. The other jobs go around the edge.
c1b_s2	Set 3: The bean bed is 8 feet long and 2 feet wide. What is its area in square feet?	**16 square feet**	2 rows of 8 is 2 × 8 = 16 square feet.
c1c_s2	Set 3: Which explanation is correct? Why is perimeter measured in feet, but area is measured in square feet?	**Perimeter is a length around the edge. Area counts the squares that cover the inside.**	Perimeter measures a length, like a fence, so we use feet. Area counts square feet that cover the inside.

SAME PERIMETER, DIFFERENT AREA

ID	Question	Answer	Why (explain to student)
d1a_s0	Set 1: Two beds both have a perimeter of 16 feet. Bed A is 1 foot by 7 feet. Bed B is 4 feet by 4 feet. Which bed has more space for planting?	**Bed B, with an area of 16 square feet**	Bed A: 1 × 7 = 7 square feet. Bed B: 4 × 4 = 16 square feet. Same fence, but Bed B holds more.
d1b_s0	Set 1: A bed is 3 feet wide and 7 feet long. Its perimeter is 20 feet. What is its area, in square feet?	**21 square feet**	Area is 3 × 7 = 21 square feet. The perimeter, 3 + 7 + 3 + 7 = 20 feet, is a different measure.
d1c_s0	Set 1: Which explanation is correct? Priya says two rectangles with the same perimeter must have the same area.	**She is wrong. A 1 by 7 and a 2 by 6 rectangle both have a perimeter of 16, but their areas are 7 and 12.**	1 + 7 + 1 + 7 = 16 and 2 + 6 + 2 + 6 = 16, but 1 × 7 = 7 and 2 × 6 = 12. Same perimeter, different areas.
d1a_s1	Set 2: Two patios both have an area of 16 square feet. One is 2 feet by 8 feet. The other is 4 feet by 4 feet. Which one needs less fence?	**4 by 4, with a perimeter of 16 feet**	2 + 8 + 2 + 8 = 20 feet and 4 + 4 + 4 + 4 = 16 feet. Same area, but the 4 by 4 patio needs less fence.

ID	Question	Answer	Why (explain to student)
d1b_s1	Set 2: Jalen has 18 square tiles. Each tile is 1 foot on each side. He makes a patio 2 tiles wide and 9 tiles long. What is the perimeter of the patio, in feet?	**22 feet**	The patio is 2 feet by 9 feet. Add all four sides: 2 + 9 + 2 + 9 = 22 feet. (Its area is 2 × 9 = 18 square feet, a different measure.)
d1c_s1	Set 2: Find the mistake. Mateo says a 1 by 8 bed and a 2 by 4 bed must have the same perimeter, because they both have an area of 8 square feet. What went wrong?	**The same area does not mean the same perimeter. The perimeters are 18 feet and 12 feet.**	1 × 8 = 8 and 2 × 4 = 8, but 1 + 8 + 1 + 8 = 18 feet and 2 + 4 + 2 + 4 = 12 feet.
d1a_s2	Set 3: Priya has 14 feet of fence. She can build a bed that is 1 foot by 6 feet or 2 feet by 5 feet. Both use all 14 feet. Which of these two beds has the bigger area, in square feet?	**10 square feet**	1 × 6 = 6 square feet and 2 × 5 = 10 square feet. The 2 by 5 bed gives the bigger area, 10 square feet.
d1b_s2	Set 3: Which two rectangles have the same area but different perimeters?	**2 by 9 and 3 by 6**	2 × 9 = 18 and 3 × 6 = 18, the same area. Their perimeters are 22 feet and 18 feet, which are different.
d1c_s2	Set 3: Which explanation is correct? Mateo has 20 feet of fence. Which bed gives more planting space: 2 feet by 8 feet, or 5 feet by 5 feet?	**5 by 5, because its area is 25 square feet and the 2 by 8 bed has only 16.**	Both use 20 feet of fence, but 5 × 5 = 25 and 2 × 8 = 16. The 5 by 5 bed holds more.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match Perimeter of a 4 by 2 bed	**12 feet**	2 points if matched on the first try.
match_s0_2	Set 1: Match Area of a 4 by 2 bed	**8 square feet**	2 points if matched on the first try.
match_s0_3	Set 1: Match Perimeter of a 5 by 5 bed	**20 feet**	2 points if matched on the first try.
match_s0_4	Set 1: Match Area of a 5 by 5 bed	**25 square feet**	2 points if matched on the first try.
match_s0_5	Set 1: Match Perimeter of a 6 by 1 bed	**14 feet**	2 points if matched on the first try.
match_s1_1	Set 2: Match perimeter	**the distance around a shape**	2 points if matched on the first try.
match_s1_2	Set 2: Match area	**the square units that cover a shape**	2 points if matched on the first try.
match_s1_3	Set 2: Match square unit	**a square that is 1 unit on each side**	2 points if matched on the first try.
match_s1_4	Set 2: Match side length	**how long one side is**	2 points if matched on the first try.
match_s1_5	Set 2: Match polygon	**a closed, flat shape with straight sides**	2 points if matched on the first try.
match_s2_1	Set 3: Match A square with a perimeter of 16 feet	**each side is 4 feet**	2 points if matched on the first try.
match_s2_2	Set 3: Match A square with a perimeter of 24 feet	**each side is 6 feet**	2 points if matched on the first try.
match_s2_3	Set 3: Match A fence around the garden	**find the perimeter**	2 points if matched on the first try.
match_s2_4	Set 3: Match Mulch to cover a bed	**find the area**	2 points if matched on the first try.
match_s2_5	Set 3: Match A 3 by 4 rectangle	**area of 12 square feet**	2 points if matched on the first try.

FIX THE SENTENCE

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: jalen built a fence around the bed	**Jalen built a fence around the bed.**	Jalen is a name and starts the sentence, so it gets a capital J. A telling sentence ends with a period.
fs2_s0	Set 1: Fix the sentence: we planted tomatoes peppers and beans	**We planted tomatoes, peppers, and beans. (also accepted: We planted tomatoes, peppers and beans.)**	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how long is the garden fence	**How long is the garden fence?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: priya mateo and jalen dug the soil	**Priya, Mateo, and Jalen dug the soil. (also accepted: Priya, Mateo and Jalen dug the soil.)**	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: the garden is in atlanta	**The garden is in Atlanta.**	Start with a capital T. Atlanta is the name of a city, so it gets a capital A. End the telling sentence with a period.
fs2_s2	Set 3: Fix the sentence: we used bricks wood and stones	**We used bricks, wood, and stones. (also accepted: We used bricks, wood and stones.)**	Start with a capital W, put commas between bricks, wood, and stones, and end with a period.

BOSS BATTLE: THE BIG GARDEN DAY

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! A rectangle bed is 9 feet long and 5 feet wide. Jalen already put up 10 feet of fence. How many more feet of fence does he need to go all the way around?	**18 feet**	Step 1: perimeter = 9 + 5 + 9 + 5 = 28 feet. Step 2: 28 − 10 = 18 more feet.
bb2_s0	Set 1: Which explanation is correct? Priya's bed is 2 feet by 12 feet. Mateo's bed is 4 feet by 6 feet. Both have an area of 24 square feet. Mateo says his bed needs less fence.	**Mateo is right. His bed needs 20 feet of fence, and Priya's needs 28 feet.**	4 + 6 + 4 + 6 = 20 feet and 2 + 12 + 2 + 12 = 28 feet. Same area, different perimeters.
bb3_s0	Set 1: The tomato bed is a square with a perimeter of 36 feet. How long is each side, in feet?	**9 feet**	A square has 4 equal sides. 9 + 9 + 9 + 9 = 36, so each side is 9 feet.
bb4_s0	Set 1: Find the mistake. Jalen found the area of a 6 by 5 bed like this: 6 + 5 + 6 + 5 = 22 square feet. What went wrong?	**He found the perimeter, 22 feet. The area is 6 × 5 = 30 square feet.**	Adding the sides gives the perimeter. Area is length × width: 6 × 5 = 30 square feet.
bb1_s1	Set 2: Two steps! The garden has a rectangle bed 7 feet by 3 feet and a square bed with 4-foot sides. How many square feet of soil will cover both beds?	**37 square feet**	Step 1: 7 × 3 = 21 and 4 × 4 = 16 square feet. Step 2: 21 + 16 = 37 square feet.

ID	Question	Answer	Why (explain to student)
bb2_s1	Set 2: Find the mistake. A rectangle bed has a perimeter of 24 feet and is 8 feet long. Mateo says it is 16 feet wide. What went wrong?	**He only took away one long side. 24 − 8 − 8 = 8, so it is 4 feet wide.**	Both long sides are 8 feet: 8 + 8 = 16. 24 − 16 = 8 feet for the two short sides, so each is 4 feet.
bb3_s1	Set 2: A garden bed has 6 sides. Each side is 3 feet long. What is its perimeter, in feet?	**18 feet**	3 + 3 + 3 + 3 + 3 + 3 = 18 feet. That is 6 × 3 = 18.
bb4_s1	Set 2: Each bed uses exactly 24 feet of fence. Which bed gives the most planting space?	**6 feet by 6 feet**	The areas are 36, 20, 32, and 11 square feet. The 6 by 6 square holds the most.
bb1_s2	Set 3: Two steps! The garden fence is a rectangle 12 feet long and 8 feet wide. The gate takes up 4 feet of one side. How many feet of fence are there, not counting the gate?	**36 feet**	Step 1: perimeter = 12 + 8 + 12 + 8 = 40 feet. Step 2: 40 − 4 = 36 feet of fence.
bb2_s2	Set 3: Which explanation is correct? Mateo's bed is 3 feet by 8 feet. What does the number 22 tell about it?	**22 feet is the distance around the bed, its perimeter.**	3 + 8 + 3 + 8 = 22 feet is the perimeter. The area is 3 × 8 = 24 square feet.
bb3_s2	Set 3: A rectangle bed has a perimeter of 30 feet. It is 10 feet long. How wide is it, in feet?	**5 feet**	The two long sides are 10 + 10 = 20 feet. 30 − 20 = 10 feet for the two short sides, so each is 5 feet.
bb4_s2	Set 3: Find the mistake. Priya says a 2 by 7 bed and a 4 by 5 bed must have the same area, because they have the same perimeter. What went wrong?	**The same perimeter does not mean the same area. The areas are 14 and 20 square feet.**	Both perimeters are 18 feet, but 2 × 7 = 14 and 4 × 5 = 20 square feet.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Area of 3 rows of 6 tiles	3 × 6 = 18 square units.
2	Distance around a 5 by 4 bed	Sides 5, 4, 5, 4. 5 + 4 + 5 + 4 = 18 feet.
3	Word for the distance around	perimeter
4	Triangle 3, 4, 5 inches	3 + 4 + 5 = 12 inches.
5	Rectangle 2 by 4: area and distance around	Area 2 × 4 = 8 square units; perimeter 2 + 4 + 2 + 4 = 12 units.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	9 by 6 bed: perimeter and area	Perimeter 9 + 6 + 9 + 6 = 30 feet; area 9 × 6 = 54 square feet.
2	Square with perimeter 28 feet	7 feet, because 7 + 7 + 7 + 7 = 28 (or 28 ÷ 4 = 7).
3	Rectangle, perimeter 26 feet, length 9 feet	Step 1: 9 + 9 = 18. Step 2: 26 − 18 = 8, split into 4 + 4. The bed is 4 feet wide.

#	Prompt	Model answer / grading note
4	Mistake: 1 by 9 and 3 by 3 need the same fence	Same area does not mean same perimeter. 1 + 9 + 1 + 9 = 20 feet, but 3 + 3 + 3 + 3 = 12 feet, so the 1 by 9 bed needs more fence.
5	Two rectangles with perimeter 12	Any two of: 1 by 5 (area 5 square units), 2 by 4 (area 8 square units), 3 by 3 (area 9 square units).

★ CHALLENGE

#	Prompt	Model answer / grading note
1	All rectangles with perimeter 16 feet	1 by 7 (area 7), 2 by 6 (area 12), 3 by 5 (area 15), 4 by 4 (area 16 square feet). The 4 by 4 square gives the most planting space.